

Abdul Basit Butt

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EDUCATION

University of Management and Technology (UMT)

2023 – 2027

B.S. Electrical Engineering

Lahore, Pakistan

Punjab Group of Colleges

F.Sc. Pre-Engineering – 70%

Gujranwala, Pakistan

A Plus School System

SSC, Biology Science – 97%

Gujranwala, Pakistan

TECHNICAL SKILLS

Programming & Development : Python, MATLAB, C++, JavaScript, HTML/CSS, Embedded C, Git, GitHub

AI & Machine Learning : TensorFlow, Scikit-learn, CNNs, Transfer Learning, EfficientNet, YOLOv8, OpenCV, GRU, BiLSTM

Electrical Engineering : Power Systems, Transformer Design, Power System Protection, Control Systems, Signal Processing, Instrumentation

Simulation & Engineering Tools : MATLAB, Simulink, Root Locus, State-Space Analysis, PSCAD, Engineering Software Development

Embedded & Real-Time Systems : STM32, Arduino, UART, Sensors, Real-Time Systems, Safety-Critical Systems

Research & Technical Communication : LaTeX, Scientific Writing, Literature Review, Experimental Design, Data Analysis, Technical Documentation

PROJECTS

Transformer Design System (TDS) | Python, Electrical Engineering

Developed a modular transformer design engine implementing electrical, magnetic, winding, insulation, and dimensional calculations, with reusable engineering modules for core, winding, window, insulation, and automated report generation, establishing a scalable foundation for automated transformer design, analysis, and future physical validation.

Transmission Line Protection with STATCOM | MATLAB, Simulink

Analyzed transmission-line distance protection under STATCOM-based FACTS compensation, evaluating mho and quadrilateral relay characteristics, protection zones, impedance measurements, and multiple fault conditions to assess the impact of STATCOM on relay performance and protection reliability.

Twin Rotor MIMO System Analysis & Control | MATLAB, Simulink

Developed and analyzed a sixth-order state-space model of the Twin Rotor MIMO System, performing stability, controllability, observability, eigenvalue, transfer function, and root locus analyses, and designed a Lead-Lag compensator achieving approximately 2.38% overshoot while maintaining closed-loop stability.

Industrial Boiler Monitoring System | MATLAB, Simulink

Designed and simulated an industrial boiler temperature and flame monitoring system using Pt100 RTD and photodiode sensors, implementing signal conditioning, ADC conversion, filtering, threshold-based detection, and over-temperature alarm logic, with validation across multiple operating conditions.

Automatic Emergency Braking System | STM32, Embedded C

Developed a real-time embedded Automatic Emergency Braking system using ultrasonic and ToF distance sensors, implementing UART communication, interrupt-driven processing, watchdog protection, and stopping-distance calculations, and validated obstacle detection, response latency, and fail-safe behavior through integration and fault-injection testing.

Vehicle Image Classification System | Python, TensorFlow, EfficientNetB0

Developed a deep learning system for classifying 20 vehicle categories using EfficientNetB0 transfer learning, achieved approximately 90% independent test accuracy, and implemented real-time image prediction through Streamlit.

Electrical Engineering Toolkit | HTML, CSS, JavaScript

Developed a web-based electrical engineering toolkit providing calculators for power systems, transformers, circuits, and related analysis, using a modular calculation architecture and responsive interface to streamline common engineering computations.

ARTICLES

An Investigation into Cross-Domain Dataset Contamination and Class Confusion in YOLOv8-Based Smart Surveillance Systems

2026

Preprint investigating sequential multi-domain retraining in YOLOv8-based smart surveillance, focusing on cross-domain misclassification, class confusion, prediction consistency, and inference stability. [Zenodo]

Artificial Intelligence Based Control Systems for Robotics and Renewable Energy Applications

2026

Co-authored a review of AI-driven control strategies, including neural networks, fuzzy logic, reinforcement learning, evolutionary optimization, and hybrid intelligent controllers for robotics, power electronics, microgrids, and renewable energy systems. [PDF]

Performance of Distance Relay with Different Impedance Characteristics in the Presence of STATCOM for Various Faults on Transmission Lines

2026

Co-authored a research paper investigating the effect of STATCOM-based FACTS compensation on transmission-line distance protection, evaluating mho and quadrilateral relay characteristics, protection zones, impedance measurements, and relay performance under various fault conditions. [PDF]

A Comparative Literature Review of Hard Disk Drives and Solid-State Drives

2026

Conducted a comparative technical review of HDD and SSD technologies, synthesizing recent literature on performance, reliability, endurance, energy efficiency, cost, and application suitability across consumer, enterprise, and cloud computing environments. [PDF]